



Varendra University
Department of Computer Science and Engineering

Undergraduate Thesis/Project Proposal

Title of the Thesis/Project
(Write your group proposal title)

Submitted by:

Name of the Student-1 (ID of the Student-1)
Name of the Student-2 (ID of the Student-2)
Name of the Student-3 (ID of the Student-3)

Program: B.Sc. in Computer Science and Engineering

Semester: 7th

Section:

Course Code: CSE 4100

Course Title: Project or Thesis with Seminar part I

Supervisor:

[Full Name]
[Designation]
[Department Name]

Submission Date: [DD Month YYYY]

1. Introduction

Introduce the topic, its background, and significance, highlighting its relevance to current technologies. Ensure that the section adequately discuss about the principles or computational frameworks to the proposed solution. (Approx. 200–300 words)

2. Literature Review / Background Study

Summarize 3–5 key existing works (research papers, systems, tools, products). Explain their contributions and clearly identify gaps or limitations that your work addresses. (Use proper citations — IEEE style recommended.)

3. Problem Statement

Describe the specific problem or challenge your thesis/project will solve. For these, this should reflect a research gap. For projects, this may involve real-world inefficiencies or limitations. (Use 1–2 focused paragraphs.)

4. Objectives

List the intended goals of your work. These should be SMART — Specific, Measurable, Achievable, Relevant, and Time-bound. (Use bullet points, typically 2–4 objectives.)

5. Proposed Methodology / System Design

Explain how you plan to achieve your objectives: For a thesis, describe your research methods, models, and/or data. For a project, define the system architecture, modules, tools, and process flow. (Include diagrams or flowcharts if applicable.)

6. Method Justification

Explain and justify why the selected methodology/system design fits the FYDP objectives/requirements.

7. Expected Outcomes

List or describe the main deliverables or findings you expect to achieve — e.g., working system, experimental results, research insights, prototype, etc.

8. Tools and Technologies to Be Used

List the software, programming languages, libraries, hardware, or datasets that will be used to implement or support your work.

9. References

Follow IEEE referencing style. Include at least 5 references.

10. Work Division and Team Management

Describe how tasks will be distributed among team members. Use a table to show task assignment. (Clarify roles.)

11. Timeline

Provide a time-based breakdown of your work (e.g., by weeks or months). Use a table or Gantt chart.

12. Cost-Benefit Analysis

Estimate any costs for tools, software, hosting, datasets, or devices. Mention if it is low-cost, cost-effective, or funded. Assess the economic feasibility, compare different options, and select the most cost-effective one. If no cost, state that clearly.

13. Constraints and Limitations

Mention only the constraints relevant to your work. These may include economic, environmental, ethical, health and safety, social, political, or sustainability-related factors that affect the design, development, or research process.

14. Societal Impact

Briefly explain how your proposed work will benefit society. This may include improvements in quality of life, accessibility, education, public health, digital inclusion, environmental awareness, or other social aspects. Focus on real-world relevance and ethical contribution to the community or specific user groups.

15. Complex Engineering Problem Mapping

Fill this table by reflecting how your proposal maps with the complex engineering problem domains.

Table 1: Mapping with complex problem solving.

WP1 Depth of Knowledge	WP2 Range of Conflicting Requirements	WP3 Depth of Analysis	WP4 Familiarity of Issues	WP5 Extent of Applicable Codes	WP6 Extent of Stakeholder Involvement	WP7 Interdependence
✓	✓	✓	✓			

16. Complex Engineering Activity Mapping

Justify which complex engineering activity level your proposal aligns with.

Table 2: Mapping with complex engineering activities.

EA1 Range of resources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓			

17. Green Computing Considerations

Mention how your work considers energy efficiency, environmental sustainability, or digital carbon footprint reduction. (Even if indirectly related, reflect awareness of green ICT principles.)

18. Lifelong Learning and Personal Development

State how this work will contribute to your knowledge, skills, and long-term professional growth. (E.g., learning new tech, teamwork, research skills, etc.)